

"Count on Me": Enabling Collaboration Through Active Support for Primary Family Caregivers in Chronic Illness Care

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Abstract

Family caregivers play a critical role in chronic illness care, taking on multifaceted responsibilities that include both medical and non-medical tasks. A variety of technologies have been developed to help caregivers manage the physical and emotional strain of their daily lives. However, existing technologies are not designed to support collaborative caregiving, which contributes to the burden on primary caregivers who often feel solely responsible. This study is part of a larger research project that includes design probes, interviews, and iterative design. In this paper, we report on an iterative design process aimed at generating design concepts, which led to two low-fidelity prototypes: 'Active Support' and 'Reflective Caregiving Tracker.' We discuss design opportunities to alleviate caregiver guilt and the burden of sole responsibility. Findings from this study will inform the next iteration of our design, which will involve a feasibility test with a large number of family caregivers and healthcare providers.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

Keywords

Family, caregiving, technology, chronic illness care, design

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1 Introduction

Family caregivers play a critical role in chronic illness care, taking on multifaceted responsibilities [1, 16]. These responsibilities often involve significant physical and emotional strain, creating challenges in their daily lives. In HCI, researchers have recognized caregivers' unique needs and social identities, suggesting technology-mediated interventions to support their work, such as providing information and emotional support [2, 12, 13, 15, 16, 19, 27–31]. For example, Chen et al. highlighted caregivers' constant efforts to balance their caregiving roles with personal responsibilities [4]. Similarly, Bhat and colleagues viewed caregivers as central to patient care and examined how they navigate their roles across different social contexts [1].

Previous studies have expanded perspectives by viewing caregivers as individuals who bring value and effort to patient care and have discussed design approaches for technologies that support them [17, 24]. However, existing interventions focus primarily on helping caregivers complete their tasks effectively, rather than alleviating their overwhelming burden, by distributing responsibilities to others, such as family members or friends. Although researchers have explored various solutions, many of these technologies add to the workload of primary caregivers. This limitation misses the opportunity for collaborative family caregiving, which could potentially have a positive impact on both caregivers and patients. Furthermore, many HCI studies on family caregiving have focused on front-end research—identifying barriers and suggesting design implications through qualitative methods such as interviews and observations [22]—but have yet to translate these findings into tangible solutions that integrate with the existing healthcare system.

This study shows how design implications for caregiver support technologies can be translated into technology-mediated solutions. These prototypes have the potential to evolve into refined products that assist family caregivers in chronic illness care, including cancer. We also present initial feedback from family caregivers of patients who have recently undergone life-threatening treatments (e.g., hematopoietic cell transplants) and propose further iterations of our design approaches. As an important first step toward developing technology-mediated solutions for collaborative family caregiving, this study lays the groundwork for future improvements.

The findings will inform the next iteration of our design, which will include a feasibility test with a larger group of family caregivers and healthcare providers.

2 Related Study: Technologies to Support Family Caregiving

In this paper, we define "caregivers" as unpaid volunteers, typically family members and friends, who provide care for their loved ones experiencing chronic conditions that limit their independence [24]. As the role of caregivers in chronic illness care has gained increasing attention, studies across various fields—including HCI, public health, and nursing—have identified factors that negatively affect caregivers' quality of life and suggested strategies to improve their caregiving experiences. For example, social isolation, resulting from various restrictions (e.g., the immunosuppressive status of patients and time constraints), has been identified as a key factor that significantly impacts caregivers' psychological health, particularly in chronic illness care [3, 12, 14]. Other studies have found that communication barriers with patients and uncontrollable situational factors contribute to emotional strain among caregivers [2, 15, 27, 31].

To address these challenges, various health interventions and services have been introduced, including social support services, online communities, mobile apps, and web-based portals [19, 21, 23, 26, 32]. These studies acknowledge the importance of family caregiving and have led to a wide range of technology-mediated solutions [4, 12, 13, 22]. Given that chronic illness care often involves complex medical information—such as interpreting test results and coordinating care with multiple specialists—these caregiver studies have mainly focused on health literacy and assistive technologies for primary caregivers. They have examined approaches to better present and deliver information to caregivers, which can directly improve patients' health outcomes [11, 16, 30]. Interventions include health education for family caregivers (e.g., [26]), decision-making support (e.g., [19]), care coordination (e.g., [32]), and communication with healthcare providers (e.g., [21]).

However, existing technologies are primarily designed to help caregivers complete their tasks, often assuming that caregivers alone are responsible for managing various duties, such as supporting patients with daily activities, assisting with self-care, and accompanying them to clinic visits. As a result, these approaches overlook the potential for collaborative caregiving, where responsibilities are shared with others, rather than being placed solely on the caregiver. Moreover, effective methods for seeking assistance and prioritizing caregivers' well-being have not yet been adequately considered or developed.

3 Methods

In this section, we describe the interview and data analysis process, as well as how we translated the identified design implications into low-fidelity prototypes. Our larger study included a two-week design probe with family caregivers, and the findings from that phase informed the development of the interview questions for the current phase. The outcomes of the design probe will be reported separately in a future publication.

3.1 Semi-structured interviews to identify main design implications

Data Collection. From September 2023 to October 2024, we interviewed 18 participants (10 primary family caregivers and 8 healthcare providers) to explore key implications for technologies designed to support family caregivers. Prior to these interviews, we conducted a design probe with family caregivers, which informed the development of the interview questions. Each interview was conducted via Zoom and lasted approximately one hour. Main themes from interviews with family caregivers included the challenges they faced, coping strategies they developed, helpful resources they identified, technologies they used for caregiving, and their communication experiences with close family members and healthcare providers. Example interview questions included: 'How have your family routines changed before and after the diagnosis?' and 'Can you walk me through how you acquired caregiving resources?' Interviews with healthcare providers focused on their practices in working with family caregivers during hospital stays, how they communicate essential information, and any barriers they observed in caregiving. The goal was to gain a comprehensive understanding of caregivers' experiences and strategies. Example interview questions for providers included: 'How did you typically help caregivers alleviate their challenges?' and 'What are caregivers' major barriers, difficulties, and concerns?'

Participants. We recruited 10 caregivers through referrals from the M Health Fairview hospital, a university-affiliated medical center. All caregivers were actively providing care for family members hospitalized for hematopoietic cell transplant (HCT) and shared real-time information about their dynamic experiences. HCT is a procedure that infuses healthy blood-forming stem cells into a patient's body to replace bone marrow that is not functioning properly. Due to the intensive and urgent nature of transplantation, patients' spouses, children, parents, and close relatives or friends are often spontaneously required to become family caregivers at the moment of diagnosis. Whether or not they have prior caregiving experience, these individuals are expected to provide emotional support, offer transportation, assist with activities of daily living and self-care (such as line changes or flushing), and accompany patients to multiple clinic visits each week. We recruited family caregivers over the age of 18 whose patients had undergone or were scheduled to undergo HCT. The median age of the caregivers was 41 years (range: 34–69). Most caregivers were female (8) and White (9). The majority were in a parental relationship with the patient (5 mothers, 1 father), while others were the patient's child (1), spouse (2), or separated partner (1). We also recruited 8 healthcare providers through word-of-mouth referrals. All had direct interactions with family caregivers in a clinical capacity. The mean age of participants was 35.25 years (range: 30–44). Most were female (7) and White (6). In terms of professional roles, the majority were social workers (5), with the remainder being 1 nurse and 2 physician assistants.

Data analysis. All interviews were recorded and transcribed verbatim for descriptive coding and thematic analysis [5]. Using NVivo, the first author coded the data sentence by sentence to iteratively identify patterns, resulting in low-level codes (e.g., self-care – while patient is occupied, inpatient – wish for a better bed to sleep). We specifically focused on factors that caregivers identified

as useful in alleviating their difficulties and information they found helpful. The low-level codes were grouped and aggregated based on similarities and differences to identify mid-level themes (e.g., self-care depends on the patient's condition, inpatient stays are often restless). From this analysis, three main design implications emerged, which guided the iterative ideation process. Throughout the analysis, the first author regularly presented updates during team meetings, where key focuses, notable themes, and potential directions were discussed.

3.2 Iterative ideation to transform main design implication into the low-fi prototype

We began the ideation process by creating quick, rough sketches to explore a range of concepts aligned with the three main design implications identified through the interviews and data analysis, as discussed in the Results. Sketching is a common practice among designers to easily capture, explore, and communicate ideas [8]. The goal was to generate a broad spectrum of ideas without the constraints that more detailed renderings or prototypes might impose [7, 8]. At this stage, we deliberately kept the process open, avoiding limitations to specific modalities, which allowed for a wide variety of potential solutions to emerge (see Figures 1–2). As a result, the ideas we generated spanned multiple modalities, including service design, digital products, physical products, and integrated solutions that combine both physical and digital features.

4 Results

We present three main design implications identified from interviews with family caregivers and providers, and how we translated these implications into ideas for caregiver support technology.

4.1 Sharing caregiving duties with others

During the interviews, all caregivers expressed feeling overwhelmed by the sense that they were solely responsible for their patients' conditions. They acknowledged that many people offered to help, and providers also mentioned that they frequently offer assistance to caregivers. However, even when help is offered and caregivers are open to accepting it, they often experience feelings of indebtedness. In receiving support, caregivers may feel compelled to reciprocate, engage in obligatory small talk, or disclose details about the patient's condition—even when they would prefer to stay calm and maintain privacy. Furthermore, to receive the necessary and effective assistance, caregivers must repeatedly articulate their needs, making the process of requesting support more exhausting and contributing to their reluctance to seek help. As a result, support from others often remains untapped, as caregivers prefer proactive assistance that does not require them to initiate a request.

To reduce the negative consequences and facilitate the process of receiving support, we developed initial ideas to help caregivers more easily share their responsibilities with others. These strategies aim to alleviate caregivers' stress by providing simple and effective ways to receive the help they need (see Figure 1).

4.2 Cultivating and reinforcing positive emotions

We found that caregivers commonly experience negative emotions, particularly due to uncertainty and a sense of losing control over the future. Later in the caregiving process, they tend to develop their own self-care strategies, such as focusing on the present moment and emphasizing the positive aspects of their situation. This shift toward self-care stems from the recognition that, in order to provide effective care for their patients, caregivers must first attend to their own needs. However, this practice remains relatively rare while patients are actively undergoing treatment, as caregivers' time for self-care is often contingent upon the patient's condition and their other responsibilities, such as caring for other family members (e.g., the patient's younger sibling) at home.

Moreover, the interview results revealed that simple acts of recognition can truly brighten a caregiver's day. Caregivers often feel unseen, so a thoughtful gesture, such as an unexpected treat or a word of acknowledgment, can make a significant difference. For example, in the pediatric unit, a mother taking care of her son shared that being called by her name by the healthcare providers, rather than simply "the parent of [patient's name]," helped her feel more personally recognized and valued: "We get a lot of like, 'Oh, hey Mom. Oh, hey Dad,' or people just turning and looking at you in the face to speak to you... so even if it was just thanking us, it wasn't, 'Thank you,' it was, 'Thank you, [Caregiver's name]'... I didn't realize how much of a difference that would make to me to be treated as a whole person versus the service that I'm providing to the patient, but it made a huge difference." This finding aligns with insights from providers' interviews, where they mentioned that being present and a good listener are strategies to alleviate negative emotions in caregivers.

Therefore, the second design implication focuses on fostering positive emotions while acknowledging the caregiver's presence, as well as the persistent worry, anxiety, and emotional turmoil that often accompany the caregiving journey (see Figure 1). These concepts include recognition of caregivers' presence in the healthcare system, empowerment through insights into caregiving expectations, and words of encouragement and reassurance.

4.3 Leveraging the right tools at critical times

In the interviews, providers described a variety of resources available to caregivers, including an onboarding all-in-one handbook, local guidance, meal deliveries by volunteers, hospital-led support groups, financial grants, therapy sessions, and more. However, caregivers reported that many of these resources were underutilized. When asked about their awareness of existing resources, many caregivers referred to the handbook they received when the patient was admitted or mentioned it was discussed by their social worker. However, caregivers often struggled to recall specific details, such as where to find the information or when to refer to it. One caregiver shared, "We have a binder here, and there's a binder there, and then there's folders in there. And it's a lot of paperwork. It's a lot of documentation. And maybe differentiating a folder for... and I know in the white binder, there is a section for the caregiver. So, I know that there's those things there." This underutilization stems from information overload, lack of awareness, and limited time or

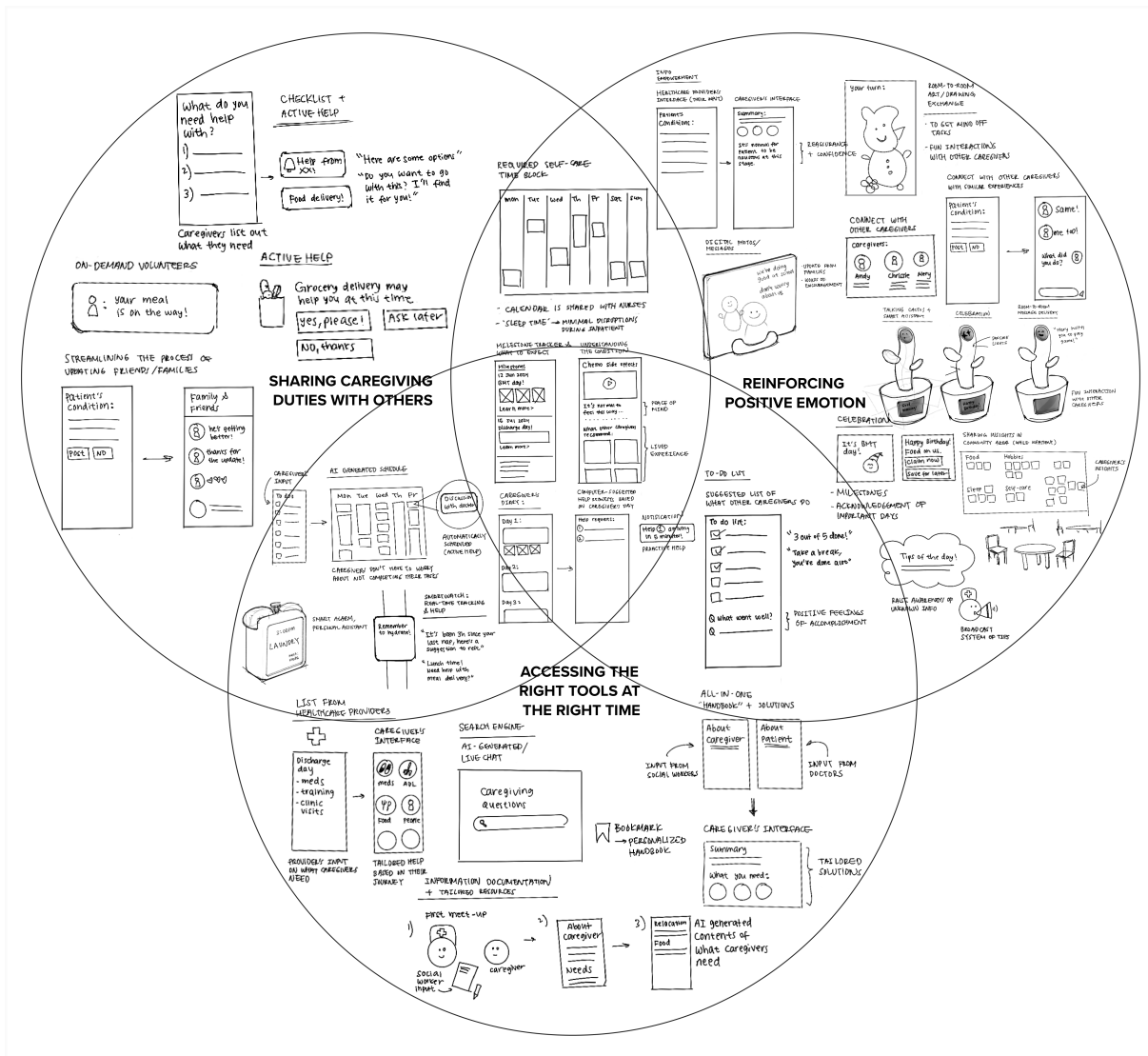


Figure 1: A diagram consisting of ideas generated.

energy to explore available options. As a result, valuable resources are often overlooked or remain underused, despite their potential to support caregivers in various contexts.

We developed the final concept to provide caregivers with adaptive resources tailored to their specific needs and circumstances (see Figure 1). From the providers' interviews, we also learned about the timeline of caregivers' involvement, including details such as when family caregivers accompany their patients to the hospital. At that time, they are required to meet with social workers and provide information to address their caregiving needs and better support the patients. Leveraging social workers' data input, we propose creating an interface where they can input relevant caregiver information into a reusable format. This could then be used for tailored resources, providing caregivers with relevant calls to

action—guiding them on what they need to know or do at any given moment—and highlighting helpful resources.

5 Low-fi Prototypes of a Smartphone-based App and User Tests

After re-evaluating all the generated ideas, we merged similar concepts, ultimately refining them into two ideas that best align with the three design criteria (see Table 1). We then created low-fidelity prototypes to visualize these two concepts and gather user feedback, a common method in the early stages of product development as part of iterative design approaches [9]. By removing color and minimizing design elements, we ensured that users' attention remained focused on the core ideas, rather than design details. This

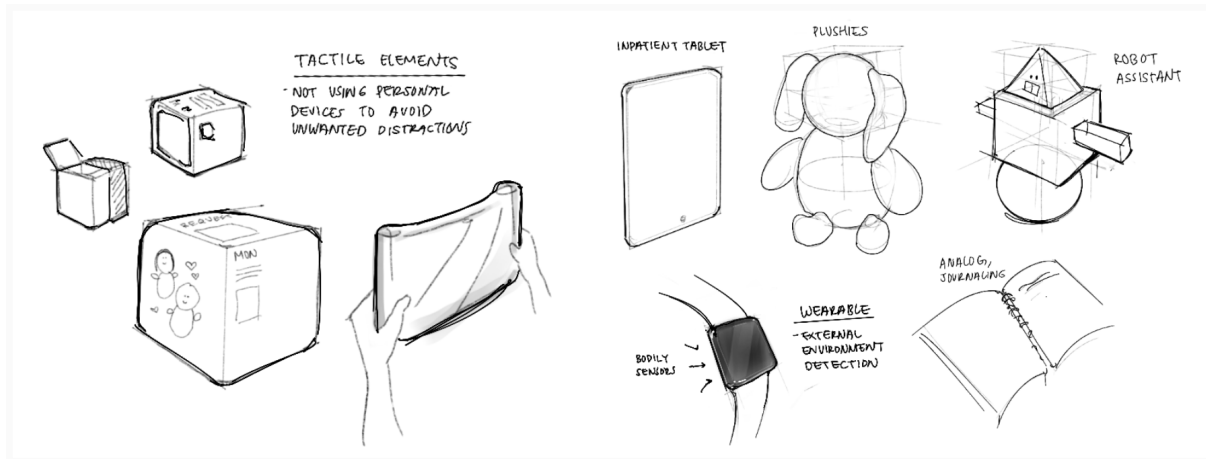


Figure 2: Exploration of various modality types.

Table 1: Main features of the two concepts that align with the design implications identified.

Design Implications	Features in Active Support	Features in Reflective Caregiving Tracker
Sharing caregiving duties with others	Creation of a custom list of their needs that is automatically shared with their private network. Utilization of information inputted by social workers to automate resource application.	The automated organization of their day, acting as their personal assistant. A collaborative calendar that can be inputted by healthcare providers for important events.
Reinforcing positive emotions	Acknowledgments of caregiving milestones, fostering positive emotions, and practical self-care strategies.	Required self-care time blocks to facilitate positive emotions.
Accessing the right tools at the right time	Provide tailored content and recommendations, depending on the stage.	Finding small chunks of time, task recommendations, including self-care activities and ongoing events.

also prevented the prototypes from appearing too polished, encouraging users to critique the ideas more freely and provide honest feedback.

We contacted caregivers from our initial interview study and conducted user tests with 3 participants who agreed to join. 2 caregivers were female and 1 was male, with a mean age of 40.67 years (range: 34–50). All participants identified as White, and all were in a parental relationship with the patient (1 father, 2 mothers).

During a 30-minute interview, each concept was introduced and reviewed one at a time. We thoroughly explained the features and functions of the concepts without discussing their potential benefits or drawbacks, maintaining neutrality to encourage honest feedback. Afterward, we asked participants how they would use the products in their caregiving tasks, when they would use them, and their perceived benefits and drawbacks. We also inquired about any additional features or content they would like to add or remove from each concept. Additionally, we presented various modalities (see Figure 2) and asked for their preferences. Below, we present the key features of the app and the main user feedback for each concept.

5.1 Active Support

The primary feature of this app is to connect caregivers with a wide range of existing resources, including institutional resources and a custom list for family and friends, while also streamlining the support request process (see Figure 3). Caregivers can create a personalized list of their needs, which can be shared with their private support network, such as family and friends who wish to contribute to caregiving. The app also compiles institutional resources (e.g., financial grants, volunteer services) and offers recommendations based on the caregiver's current phase of caregiving. Caregivers and social workers typically meet around the patient's admission date to document important details and identify care needs. This platform could also serve as a tool for social workers to input relevant caregiver information directly into the app, automating the caregiver's resource search and application process. Caregivers can select their current caregiving phase (e.g., inpatient or outpatient), as well as the patient's symptoms, and track their own emotions and stressors. Based on this input, the app will provide on-demand, personalized information about the patient and share insights from

other caregivers, giving the caregiver greater control and immediate access to relevant information whenever needed (see Figure 3). Additionally, caregivers have the option to explore other caregiving phases to better understand the typical patient journey and caregiving expectations. The app will also visually and textually celebrate milestones as caregivers progress through each stage of caregiving.

Sharing chores with family and friends to improve coordination of help. Caregivers expressed enthusiasm for the idea of sharing caregiving tasks with family and friends to improve coordination and support. They particularly appreciated the ability to create custom lists of tasks and share them with their private network. The types of tasks caregivers would include ranged from cleaning the house and babysitting to maintaining the garden to prevent the grass from growing too high—especially useful when they had to relocate with the patient. Members of their private network could then sign up to complete the listed tasks. While all caregivers found this feature helpful, some offered critiques. They shared that reaching out to family and friends can be difficult, and that a sense of pride often prevents them from seeking support. One suggestion was to structure the request in a way that makes it appear as though the task originates from the system, rather than directly from the caregiver, to reduce the perceived burden on the caregiver.

Appreciating the convenience of a centralized resources hub. When we asked caregivers about their use of or familiarity with existing resources, they shared that they do not utilize them. Caregivers noted that finding the necessary information can be both challenging and time-consuming. Although relevant resources may be available, issues such as mismatched eligibility criteria, conflicting schedules, and inconvenient locations often render them unsuitable, leading caregivers to stop referring to them. Additionally, one caregiver suggested including resources not only for caregivers, but also for patients and the entire family unit: “It was a really stressful time for me and my husband, and then also, we have these other children too. So just being able to have some kind of support for siblings—like someone they can talk to or maybe other siblings who are going through the same thing, where they can just hang out together, talk, play, and work through some stuff in their own kid way—you know, that was kind of lacking, too.” This desire aligns with our interview findings, where caregivers reported juggling multiple roles, with their well-being closely tied to the patient’s health, the overall family dynamic, and the various responsibilities they manage.

Using resources in collaboration with social workers. Caregivers appreciated the idea of involving social workers in the back-end of the application process. This not only eliminates the need for caregivers to manually enter their information when applying for resources, but it also fosters trust, making caregivers feel more comfortable with the process. During the interview, social workers acknowledged that applying for resources can be time-consuming and mentioned that they advise caregivers to apply, but caregivers reported that it is not always easy to reach a social worker. As a result, many resources may go unused. Caregivers expressed support for integrating social worker consultations into the process to ensure that these resources are fully utilized.

5.2 Reflective Caregiving Tracker

The goal of this concept is to help caregivers structure their days, ensuring they have dedicated time for tasks while also allowing them to be present and focused in the moment. The app will allocate time for self-care, normalizing it as part of the caregiving process and helping to reduce the guilt caregivers often feel when taking time for themselves. Given the unpredictability of caregivers’ schedules, the system allows them to skip tasks, with the schedule automatically updating accordingly. The reflection function is designed to be completed at the end of the day, encouraging caregivers to reflect on their experiences and identify positive aspects to enhance their well-being (see Figure 3). If caregivers skip a task, the system allows them to update the content by noting the task they completed instead. This feature acts as a personal assistant, inputting recommended tasks and helping caregivers prioritize and adjust their schedules. The collaborative calendar allows providers to input time blocks, such as doctor visits and ongoing events. This ensures caregivers can stay present without worrying about other tasks, while also keeping everyone aligned. Providers will be aware of key times, such as when caregivers are resting, and can minimize disturbances. The daily documentation of events and emotions offers caregivers an opportunity to reflect on and track their journey, as many caregivers shared in interviews that they enjoy documenting their experiences and revisiting them over time.

Reflection and thoughts to relieve stress. The reflection function was viewed as empowering, allowing caregivers to focus on the positive moments of their day and receive encouraging words. Caregivers suggested incorporating uplifting thoughts to help them unwind, reflect, and carry a sense of calm into the night. One caregiver compared this feature to the mental health app they currently use, recommending the inclusion of brief notifications with positive affirmations or self-care reminders. All caregivers preferred the phone application format for stress-relieving features due to its frequent use and ease of accessibility.

Caregivers’ preferences for using existing platforms to simplify scheduling. Many caregivers expressed limited enthusiasm for the reflective caregiving feature, noting that they already had their own strategies for managing their day, such as using other calendar platforms or developing personal routines. While they acknowledged that the feature could be beneficial for caregivers who prefer more structure to make their day more manageable, they felt that this benefit did not align with the preferences of the majority of participants.

Ongoing events can sometimes be easily missed. Caregivers appreciated having access to ongoing inpatient events inputted by providers, especially since such events are typically shared through paper flyers or word of mouth—both of which can be easily overlooked. In addition to caregiver-focused events, they also suggested including information about ongoing events for patients. This feedback aligns with our interview findings, where caregivers noted that they are more likely to engage in self-care and feel at ease when patients are doing well or are occupied.

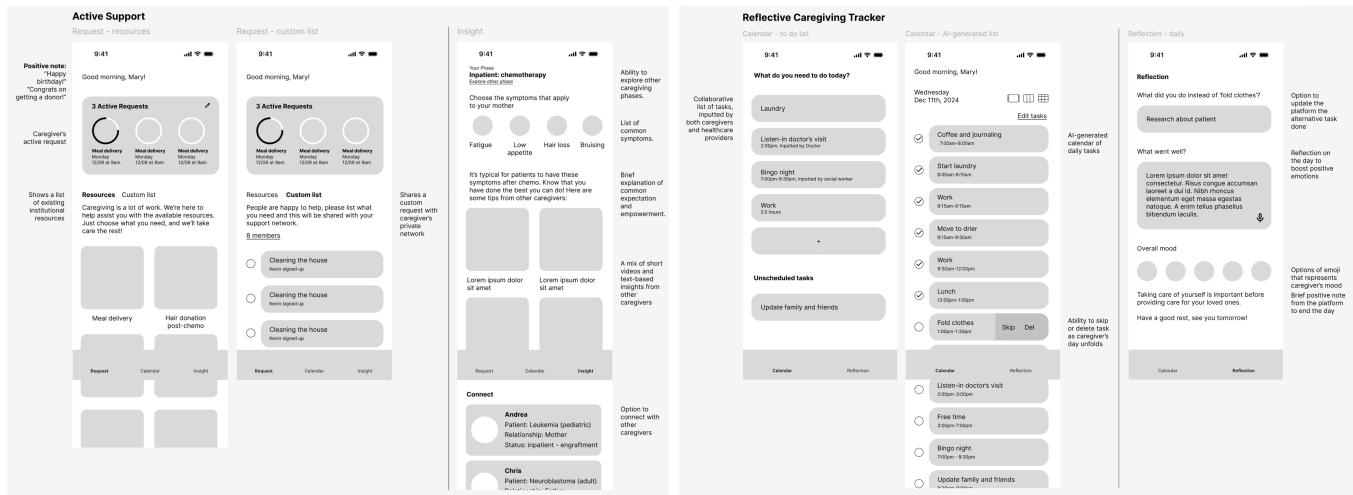


Figure 3: App Prototypes - Active Support and Reflective Caregiving Tracker.

6 Discussion and Future Work

In chronic illness care, existing technologies have primarily focused on providing more information or facilitating data tracking for primary caregivers who bear sole responsibility for managing patient care. Under current circumstances, it is challenging for family caregivers to easily divide tasks with their family and friends, as explaining and transferring responsibilities requires additional effort.

Our design enables caregivers to share their needs with family and friends, allowing them to receive necessary support without having to personally request help, thus alleviating feelings of indebtedness. However, even when the request is not directly made by caregivers, they may still feel uncomfortable, as the help request may appear to originate from them. One suggestion is to ease this discomfort by using language generated by the system or adjusting the structure of the request. This way, caregivers would feel less directly responsible for the request, allowing it to appear as though the request is coming from the platform itself or the social worker, rather than from the caregivers, which could help alleviate these uncomfortable feelings.

During the user tests, we found that caregivers expressed a strong desire for support in managing other responsibilities, such as caring for family members (e.g., young children) at home while also caring for their patients. In HCI, the family is considered a unit of support, particularly in managing everyday health, including sleep [10, 11, 20, 25]. Therefore, future efforts should focus not only on supporting individual caregivers but also on considering the caregiver’s family unit as a whole when developing caregiver support technologies. For example, providing information on ongoing events and available resources that support not just the caregivers, but also the patients and other family members, could be valuable. Additionally, exploring tasks that can be easily supported (e.g., grocery shopping, pet sitting) and those that cannot be easily handled by family and friends (e.g., assisting with self-care for patients or medication management) could significantly alleviate the burdens on primary caregivers.

Moreover, while a key feature of the active support concept is the automation of resource applications, this may require social workers to fill in missing information in the background, potentially overburdening healthcare providers with additional tasks. One possible solution could be to leverage Large Language Models (LLMs), using the contents of existing caregiver handbooks provided by hospitals as a training dataset [6]. This approach would enable the platform to automate much of the process without placing excessive demands on healthcare providers. Future work should, therefore, focus on striking a balance between automated and tailored resources for caregivers, while managing the demands placed on healthcare providers.

Our study has several limitations. First, although caregivers supported the integration of information from the patient portal, concerns about sharing protected health information (PHI) under the Health Insurance Portability and Accountability Act (HIPAA) prevented us from pursuing this integration [18]. Additionally, while caregivers can share milestones and celebrations with the care team to receive personal acknowledgment, this feature may raise ethical concerns regarding PHI. Future efforts should explore ways to protect patient health information while maximizing caregivers’ well-being. Moreover, most of our participants were White Americans from upper-middle-class backgrounds. Given existing health disparities, particularly among immigrants and non-English speakers, there is a critical need for services in other languages.

7 Conclusion

In this paper, we present an iterative design process focused on generating design concepts, which resulted in two low-fidelity prototypes: ‘Active Support’ and ‘Reflective Caregiving Tracker’. We explore design opportunities to alleviate caregiver guilt, reduce the burden of sole responsibility, support care for the entire family unit, and balance the distribution of caregiving tasks without overburdening primary caregivers or healthcare providers. The insights from this study will inform the next phase: a deployment study aimed at evaluating the feasibility of our system.

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